

# Large Language Models

*LLMs: Foundations, Generation & Beyond*

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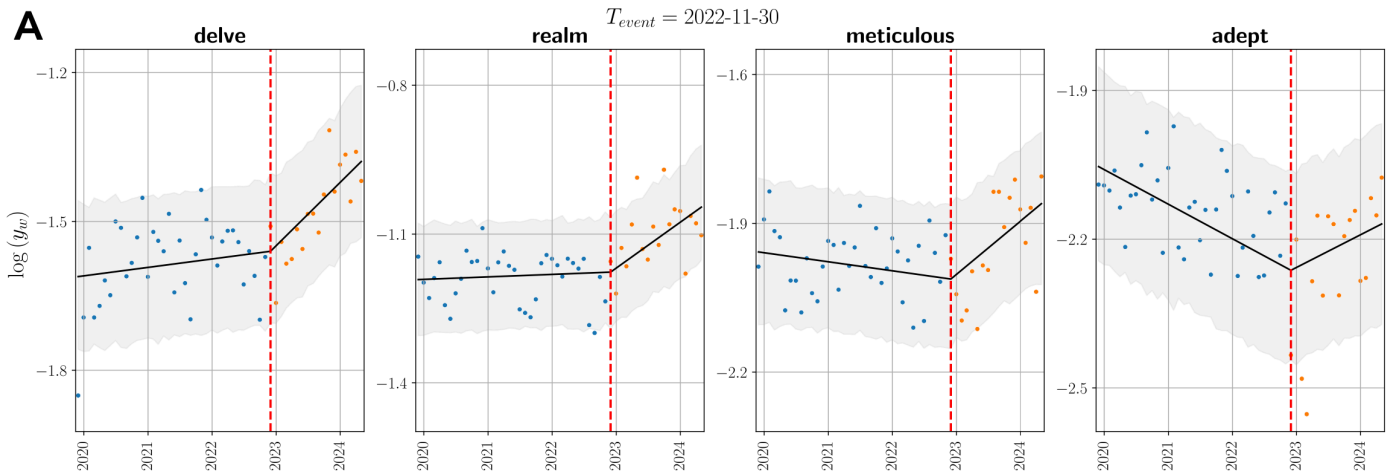
Introduction to NLP — MSc. DH EdC-PSL

November 18, 2025

**...in the real world**

# 0.1 Evidences of ChatGPT's impact

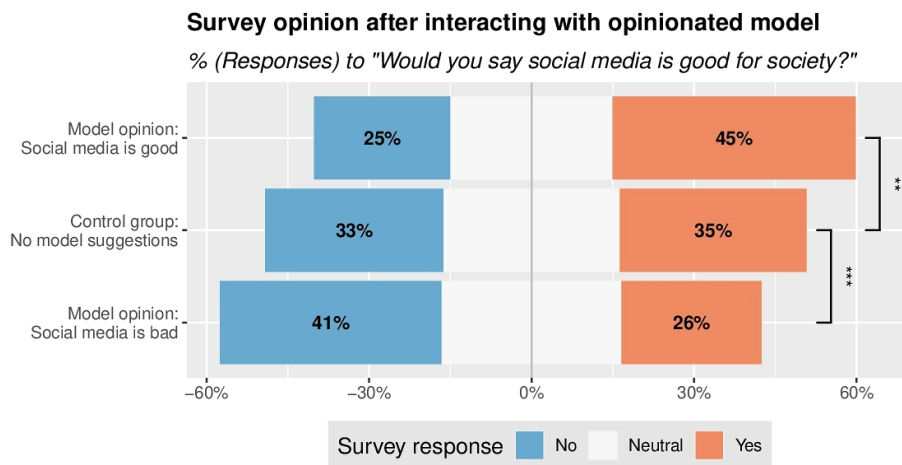
LLMs influence the way we communicate ([Yakura et al. 2024](#); [Geng et al. 2024](#); [Anderson, Shah, and Kreminski 2024](#)).



Prevalence of words associated with ChatGPT. Extracted from ([Yakura et al. 2024](#)).

## 0.2 Influence besides vocabulary

- LLMs influence users and participate in shaping opinion



- collective opinion ([Li et al. 2023](#))
- individual opinion ([Jakesch et al. 2023](#); [Williams-Ceci et al. 2024](#))

*Participants interacting with a model supportive of social media were more likely to say that social media is good for society in a later survey (and vice versa) — from Jakesch et al. (2023).*



## 0.3 Propagation of models' biases

Language is loaded with sociocultural characteristics ([Jiang 2000](#)).

→ LLMs biases can have *harmful consequences*, and *spread* in downstream applications:

- lower quality service to minorities and vulnerable users ([Poole-Dayana, Roy, and Kabbara 2024](#); [Cunningham et al. 2024](#))
- demographic-based recommendations ([Salinas et al. 2023](#))
- mental health issues and cultural values ([Hadar-Shoval et al. 2023](#))
- politically biased text summarization ([Yuhan Liu et al. 2024](#))
- ...

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# 1 Framing Large Language Models

## 1.0.1 Definition(?)

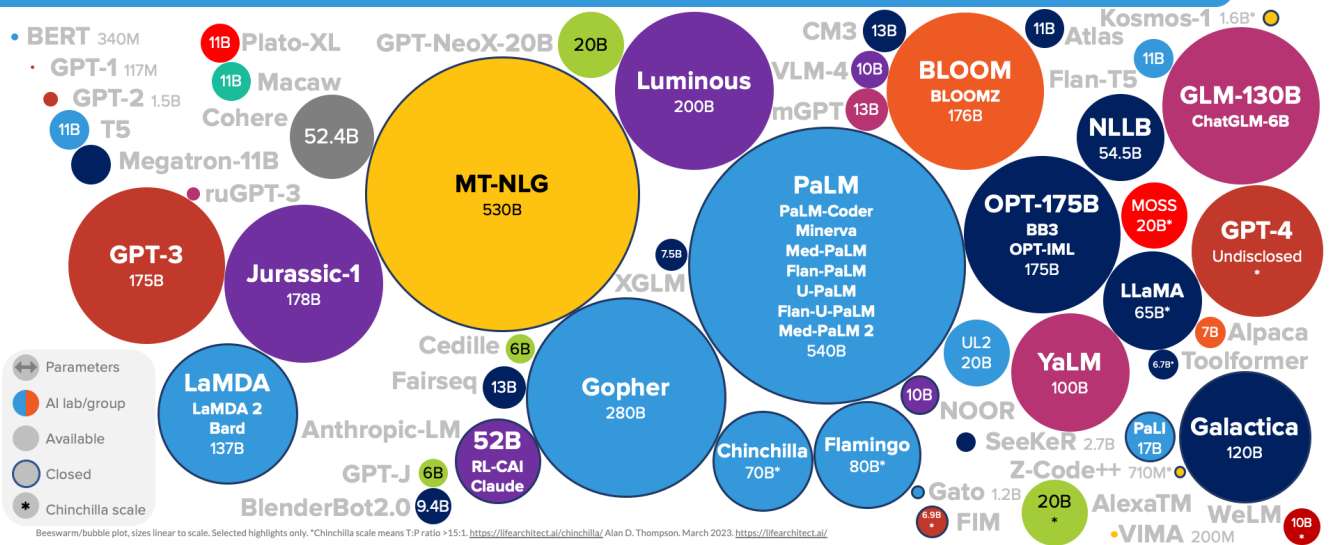
 Large Language Model - from ([Merriam-Webster 2025](#))

A language model that utilizes deep [learning] methods on an extremely large data set as a basis for predicting and constructing natural-sounding text.

## 1.1 Large

## 1.1.1 What is a *\*Large\** Language Model?

### LANGUAGE MODEL SIZES TO MAR/2023



LifeArchitect.ai/models

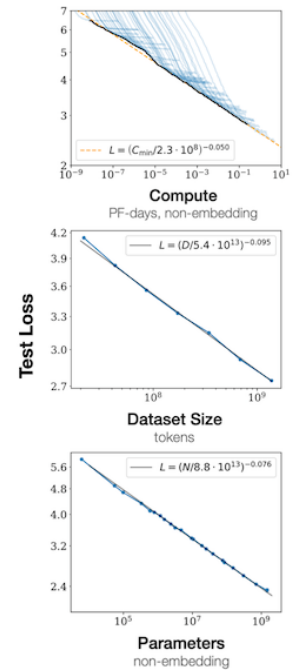
Extracted from (Thompson 2025).

# 1.1.2 Scaling Laws & The bitter Lesson

The bitter Lesson ([Sutton 2019](#))

*“general methods that leverage computation are ultimately the most effective, and by a large margin”...*

- **Knowledge** integration helps improve models in the **short term** (rewarding to researchers, but plateaus and can even inhibit further progress)
- **Breakthrough** progress eventually arrives by an opposing approach based on **scaling computation** by *learn* and *search*



Language modeling performance improves smoothly as the model size, dataset size, and amount of compute used for training increase (extracted from [Kaplan et al. 2020](#)).

## 1.2 Causal



## 1.2.1 Autoregressive

### Definition

An autoregressive (AR) model is a representation of a type of stochastic process. It specifies that the output variable (or current state) depends linearly on its own  $p$  previous values:

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t.$$

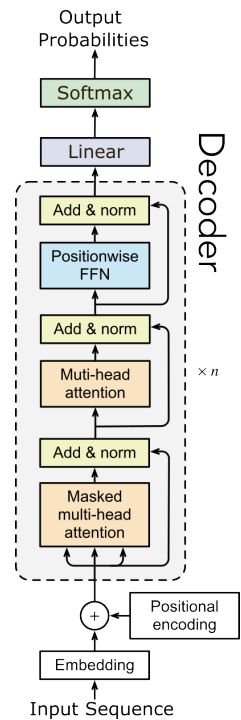
### In NLP

In NLP, AR models generally refer to *next-token* prediction models, and the linear assumption can be disregarded.

→  $n$ -grams  $\in$  LLMs ??

## 1.2.2 Decoder-only

- Only a decoder stack (no encoder)
- Causal / masked self-attention
  - Each token can only attend to *previous* tokens
- Naturally suited for **generation**
  - Trained for next token prediction
  - Produce text left-to-right, one token at a time



Transformer Architecture, based on Vaswani (2017), reworked from (Zhang et al. 2023).

## **1.3 Language Models**

## 1.3.1 The Language Modeling Problem (again)

### Language Model

A Language Model (LM) estimates the probability of pieces of text. Given a sequence of text  $w_1, w_2, \dots, w_S$ , it answers the question:

**What is  $P(w_1, w_2, \dots, w_S)$ ?**

→ encode:

- grammaticality
- semantic plausability
- stylistic consistency
- knowledge (?)

How to compute  $P$ ?

- ~~Count based (count): n-gram~~
- Neural method (learn): Transformer
  - ~~Masked Language Modeling (encoder-only, BERT)~~
  - Causal Language Modeling (decoder-only, GPT)

Large (? enough) parameters + data

~~Autoregressive~~ Causal

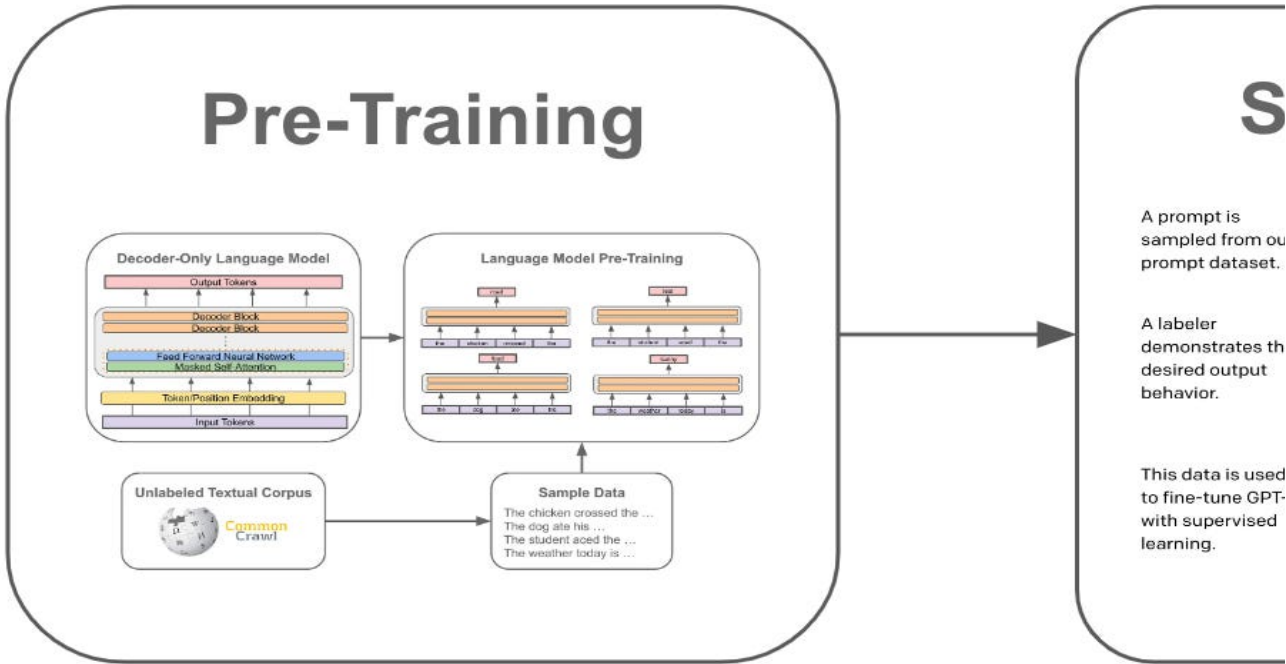
as in (→ ... yes but ... decoder-only)

Language Models

(more or less)

# 2 Training

# 2.1 LLMs training workflow



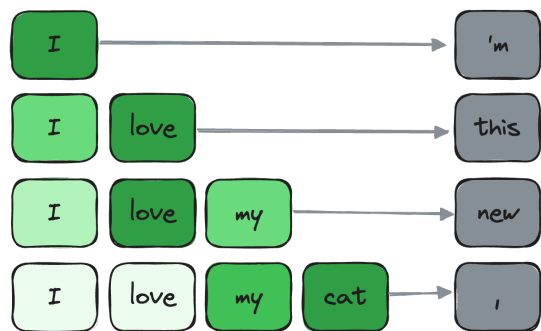
Overview of LLMs Training Pipeline. Extracted from (Wolfe 2023), based on (Ouyang et al. 2022).

# 2.2 Pre-training

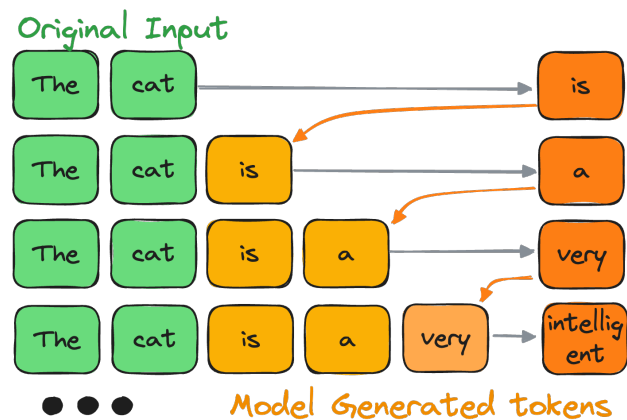


# 2.2.1 Next token prediction task

$$P(x_{i,t+1} | x_{i,1}, \dots, x_{i,t})$$



Example generated with allenai/OLMo-2-0425-1B (e.g., training).

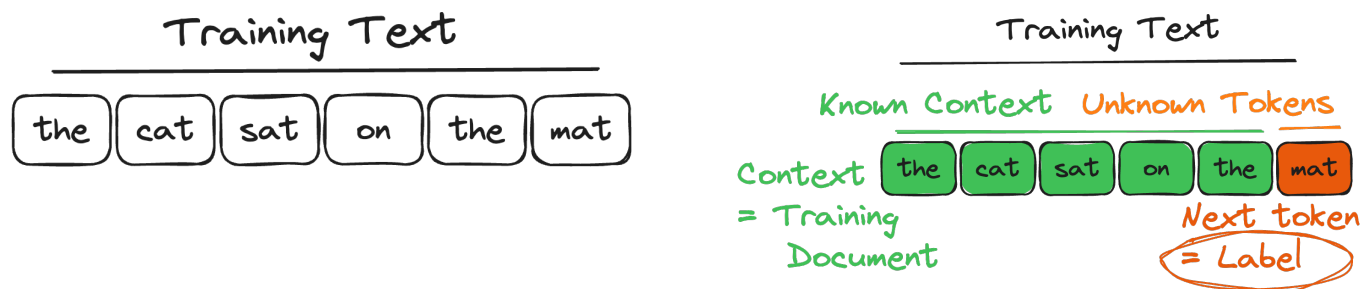


Example generated with allenai/OLMo-2-0425-1B (e.g., generation).

# 2.2.2 Self-Supervision

*i* Self-Supervised Learning (Wikipedia 2025)

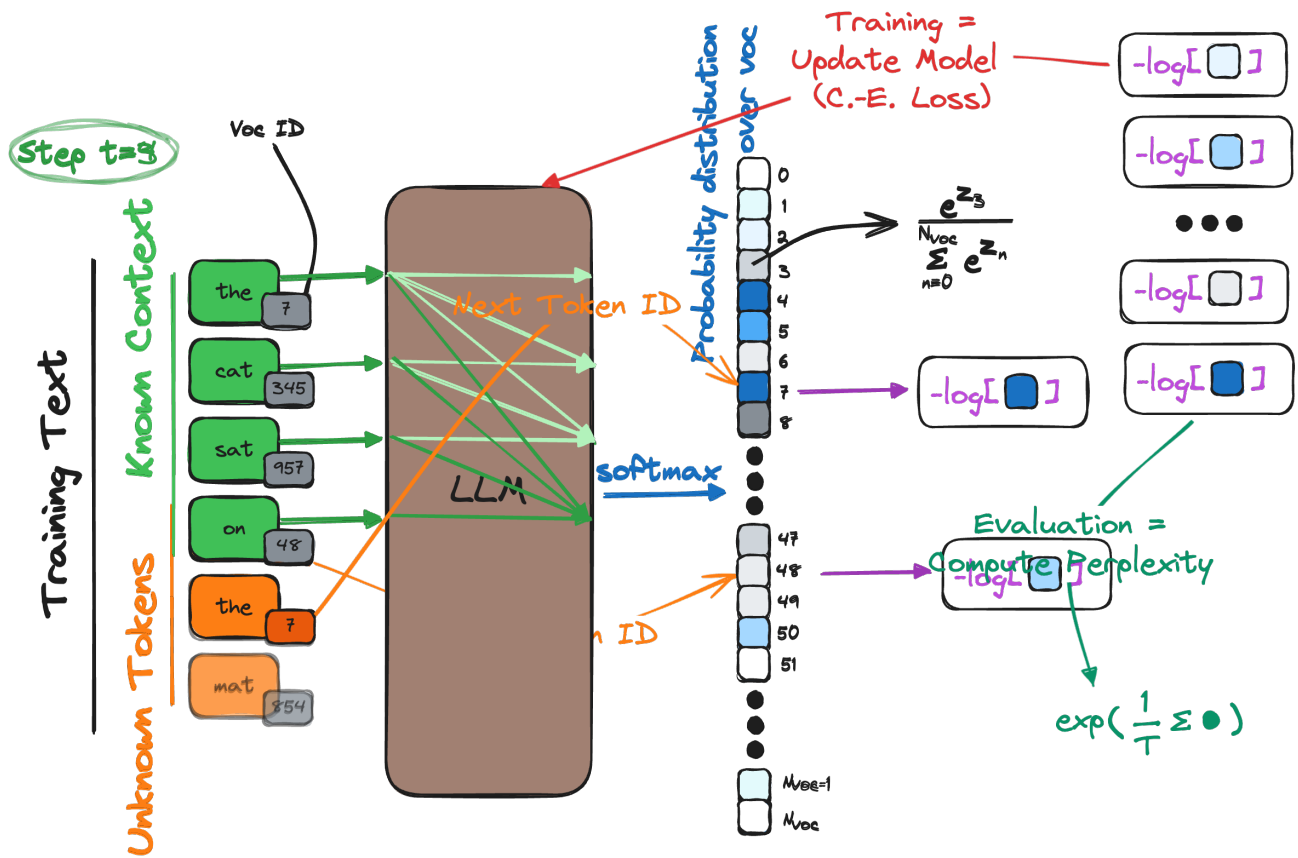
Self-supervised learning (SSL) is a paradigm in machine learning where a model is trained on a task using the **data itself** to generate **supervisory signals**, rather than relying on externally-provided labels.



## 2.2.3 Training Objective: Cross-Entropy

$$\mathcal{L}[\phi] = - \sum_{i=1}^I \sum_{t=1}^T \log [P(x_{i,t+1} | x_{i,1}, \dots, x_{i,t}, \phi)]$$

- Penalizes confident wrong predictions heavily
  - Low probability to 'true' next token  $\Rightarrow$  high loss:  $\lim_{p_i \rightarrow 0} (-\log(p_i)) = \infty$
- Rewards accurate predictions
  - High probability to 'true' next token  $\Rightarrow$  low loss:  $\lim_{p_i \rightarrow 1} (-\log(p_i)) = 0$
- Maintains Differentiability
  - smooth mathematical properties allow effective gradient-based optimization

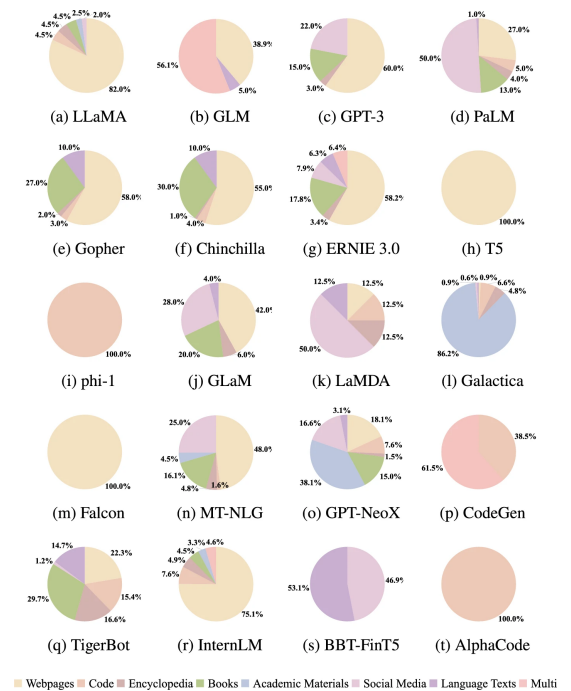


# 2.3 Data

## 2.3.1 Data Sources

Pre-training  $\approx$  *learning from the distribution of texts on the internet.*

- Primarily large-scale web crawls
  - Common Crawl, C4, CCNet, Dolma, RefinedWeb, RedPajama
- Curated “high-quality” reference works
  - Wikipedia, books, textbooks, scientific papers
- Code repositories
  - GitHub, StackOverflow



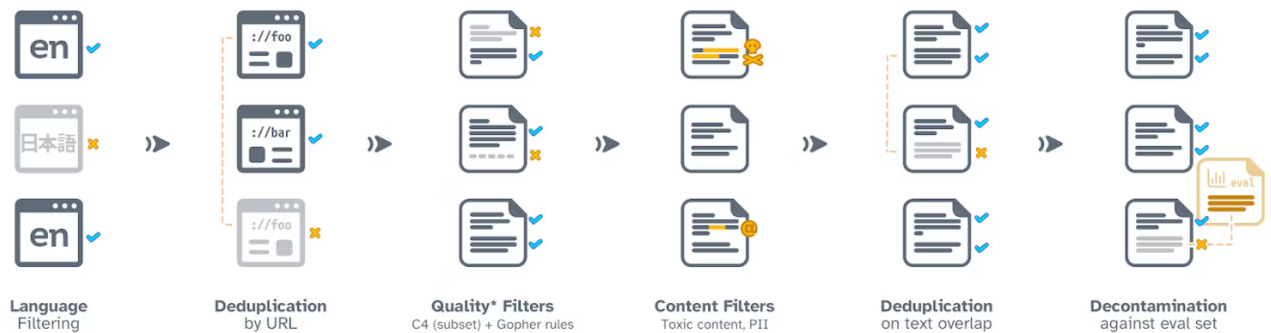
The distribution of data types in pre-training corpora used by different LLMs. Extracted from Yang Liu et al. (2025).

## 2.3.2 Data Filtering

Is that so?

```
1 python train_my_llm.py --dataset=extremely_large_dataset
```

Not *that* simple...



Overview of web data processing pipeline for Dolma (Ai2). Extracted from ([Soldaini 2023](#)).

## 2.3.3 *Quality* data?

“Quality” filtering choices...

- Monolingual and high-resource language bias ([Wenzek et al. 2019](#))
- Academic / Formal-style bias ([Gururangan et al. 2022](#))
- Textbook-like datasets ([Gunasekar et al. 2023](#))
- Wikipedia-heavy corpora ([Touvron et al. 2023](#))
- Topic / socio-economic bias ([Lucy et al. 2024](#))

... have consequences!

- Marginalization of low-resource languages, dialects, sociolects
- Overrepresentation of techno-scientific, Western, elite knowledge
- “Highbrow” epistemic bias: formal, rational, encyclopedic styles favored
- Missing registers: oral, vernacular, informal, multilingual mixing, community languages
- Potentially distorted representations of social groups or cultural practices

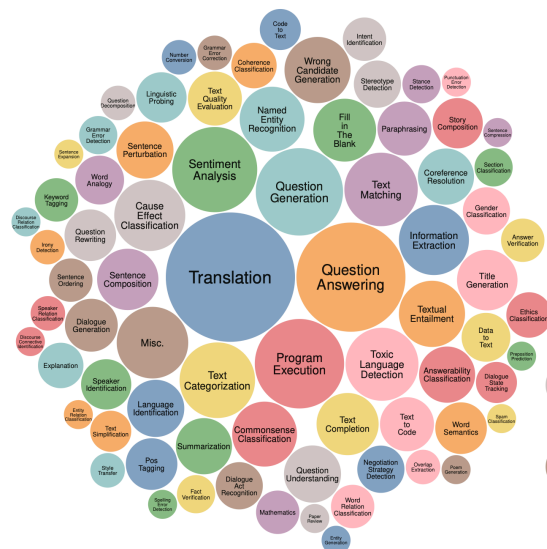


## 2.4 Post(-pre)-training

### 2.4.1 Supervised (Instruction) Fine-Tuning

**Goal:** Teach model to *follow instructions*.

- Supervised setting
  - instructions & responses
- Uses curated **input** → **output** examples
  - e.g., “Summarize...”, “Translate...”, “Explain like I’m 5...”

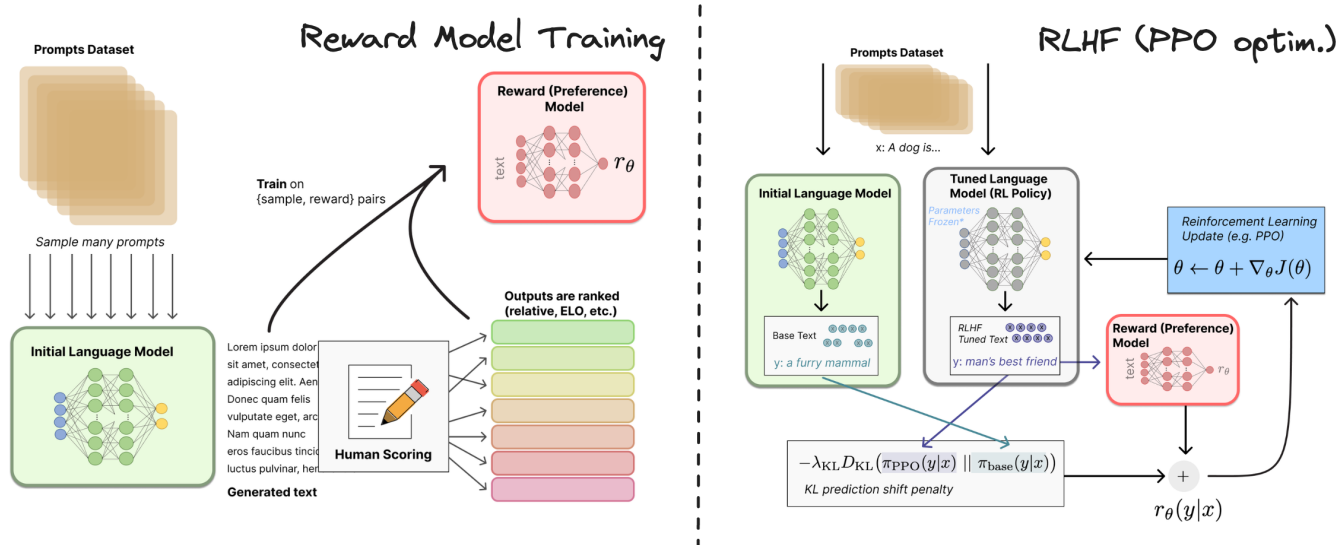


(a) SUP-NATINST (this work)

Task categories present in Super-NaturalInstructions dataset (Wang et al. 2022) (extracted from original article).

## 2.4.2 RL (from Human Feedback)

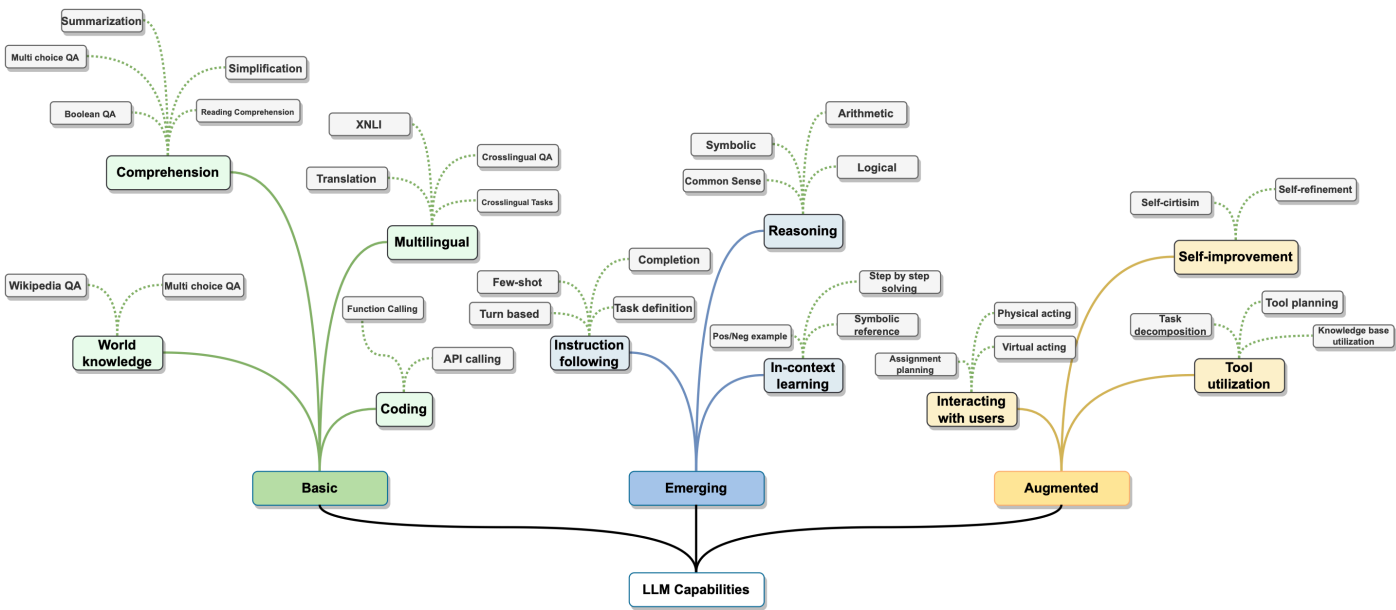
Goal: Align model with *desired behaviors* (see e.g., (Bai et al. 2022; Ouyang et al. 2022)).



Extracted from (Lambert et al. 2022).

# 3 Inference → Tutorial

# 3.0.1 Use-cases



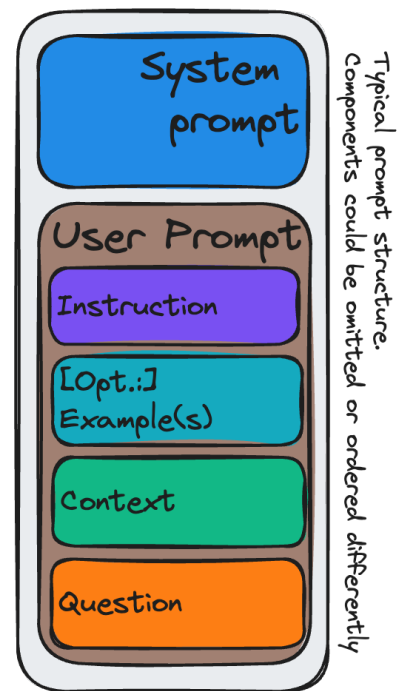
LLMs Capabilities. Extracted from ([Minaee et al. 2025](#)).

# 3.1 Prompt Strucutre

## 3.1.1 Elements of a Prompt

### Natural Language Interaction:

- System prompt
  - set of instructions, guidelines, and contextual information provided to AI models before they engage with user queries.
- User prompt
  - actual message of the user
    - Instruction: definition of the task
    - Examples: provide examples (as in few-shot)
    - Context: provide additional context to address the problem
    - Question: prompt the model to solve the task



# 3.1.2 Example: Under the Hood

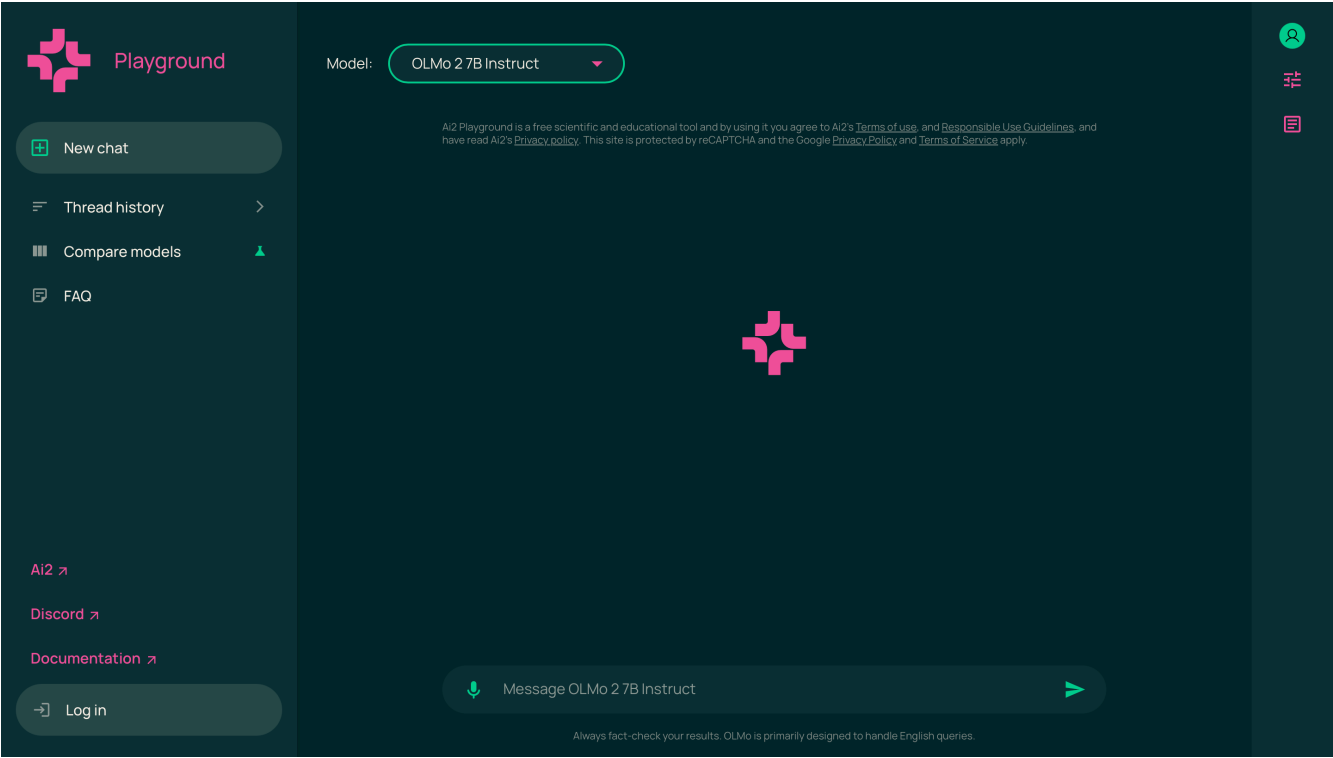
UI

Prompt

Code

Multi-turn

Multi-turn code



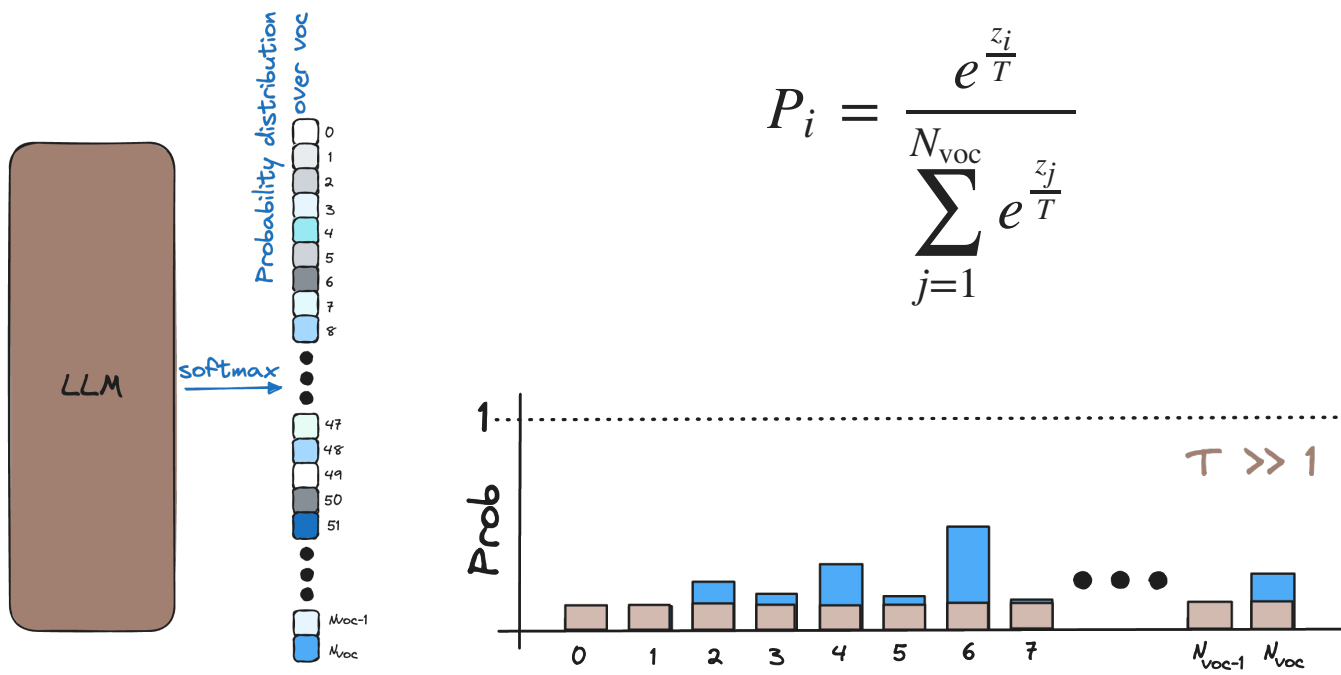
Screenshot of OLMo playground (<https://playground.allenai.org/?model=cs-OLMo-2-1124-7B-Instruct>)



3.1.3 Nice visualization [here](#) (Cho et al. 2024):

## 3.2 Sampling

# 3.2.1 Next-token probability distribution



## 3.2.2 Some Other Generation Parameters

- **temperature**

- Controls “*creativity*” vs. *determinism*.
  - ↓ low (0–0.7): predictable, focused, factual
  - ↑ high (1.0+): diverse, surprising, risk of incoherence

- **top-k**

- Keep only the **k most likely** next tokens.
  - small k → conservative, avoids rare tokens
  - large k → more expressive, less stable

- **top-p (nucleus sampling)**

- Keep the **smallest set** of tokens whose cumulative probability  $\geq p$ .
  - $p \approx 0.0$ : only most likely token
  - $p \approx 0.9$ : diverse but meaningful
  - $p \approx 1.0$ : no filtering

- **length\_penalty**

- Encourages longer or shorter outputs.
  - $> 1$ : longer answers
  - $< 1$ : concise answers

- **repetition\_penalty**

- Penalizes reusing tokens/sequences.
  - $> 1.0$  reduces loops and obsessive repetition

- **no\_repeat\_ngram\_size**

- Hard constraint: forbids repeating n-grams of size  $n$ .

# 4 Evaluation

## 4.1 Challenges

## 4.1.1 What's different?

LLMs are stochastic **generative** <sup>(~intrinsic)</sup>,

- natural language quality (fluency, coherence, relevance)
- correctness / factuality
- robustness
- safety / toxicity

**general purpose** models <sup>(~extrinsic)</sup> ([Radford et al. 2019](#)).

- general knowledge
- task-specific abilities
- calibration & reliability
- societal impact (bias, fairness, harms, ...)

→ **no single metric** can conveniently evaluate LLMs capabilities and risks ([Liang et al. 2023](#); [Bommasani et al. 2022](#))!

## 4.1.2 What to evaluate?

LLM evaluation is typically organised around (following and adapted from (Guo et al. 2023; Chang et al. 2024)):

- **NLP Abilities**

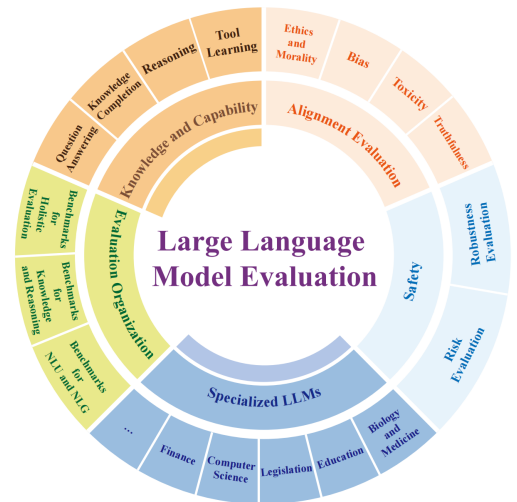
- NLU, NLG, fluency, coherence, grammaticality, ...

- **(Specific) Knowledge and Abilities**

- QA, reasoning, classification, ...
- Domain specific, e.g.: education, finance, health, ...

- **Safety and Social Impact**

- misinformation, bias, representational harms, disparate treatment, ...



*Proposed taxonomy of major categories and sub-categories of LLM evaluation, extracted from (Guo et al. 2023).*



## 4.1.3 Evaluation Approaches (overview)

- Automated
  - metric-based (perplexity, accuracy, BLEU, etc.)
  - scalable, reproducible
  - limited for open-ended tasks
- Human
  - comparative assessment
  - qualitative insights
  - “vibe check”
- Benchmark-Based
  - standardized datasets & tasks
  - often MCQA or constrained formats
  - useful for comparison but may not generalize
- Adversarial
  - robustness against prompts, attacks, edge cases
  - surfaces safety issues and model brittleness

# 4.1.4 Leaderboard Example (Fourrier et al. 2024)

[Leaderboard](#) [Citations](#) [Compare models](#) [About](#)



## Open LLM Leaderboard Archived

Comparing Large Language Models in an open and reproducible way

Search by model name • try "meta @architectur"4576 / 4576

Advanced Filters

Supports strict search and regex • Use semicolons for multiple terms

Quick Filters

For Edge Devices • 786

For Consumers • 430

Mid-range • 3185

For the GPU-rich • 165

☐ Only Official Providers • 470

## 4.2 Machine Behavior

## 4.2.1 Machine Behavior ([Rahwan et al. 2019](#))

“[Machine Behavior](#) is concerned with the scientific study of intelligent machines, not as engineering artefacts, but as a class of actors with particular behavioural patterns and ecology. This field overlaps with, but is distinct from, computer science and robotics. It treats machine behaviour empirically. This is akin to how ethology and behavioural ecology study animal behaviour [...]”

See also *Machine Psychology* and related literature, e.g., ([Bommasani et al. 2022](#); [Binz and Schulz 2023](#); [Hagendorff et al. 2024](#);

[Ye et al. 2025](#)).

## 4.2.2 Goals

- Systematic assessment of behaviors
- Studying ML-systems as a “class of actors with particular behavioural patterns and ecology” / socio-cultural artefacts
  - Which kinds of (human-relevant) capabilities are present?
  - What do they tell about the models?
- Using experimental methods inspired from behavioral psychology ([Ye et al. 2025](#)) and HSS ([Bommasani et al. 2022](#))

## 4.2.3 [Side note:] Relation to *bias*?

### Bias (definition attempt)

**Systematic** tendency in model outputs or behaviors that *produce, reflect* or *amplify* *disparate, unfair* or *harmful treatments* of individuals or social groups ([Barocas and Selbst 2016](#)). This encompasses broad ranges of discriminative patterns, *inter alia*: allocational harms, representational harms, stereotyping and unequal system performances, or questionable correlations ([Bender et al. 2021](#); [Blodgett et al. 2020](#)).

### Caution

This is a definition attempt. In NLP, the notion of “bias” is often used as an (ill-defined) umbrella term covering diverse ranges of “biased” behaviors. Blodgett et al. ([2020](#)) call for more rigorous contextual and operational definitions.

## 4.2.4 Examples: MCQ-based Evaluation

- Political Leaning

- Political Compass Test (PCT)\*

Sex outside marriage is usually immoral.

- Cultural norms and values

- World Value Survey (WSV)\*
- Global Attitude Survey

When jobs are scarce, employers should give priority to people of this country over immigrants.

- Moral values and personality-like traits

- Moral Foundations Questionnaire (MFQ)
- Myers Briggs Type Indicator
- Big-5\*

You make friends easily.

\*: Example shown.

## 4.2.5 Examples: MCQ-based Results

- Political Leaning

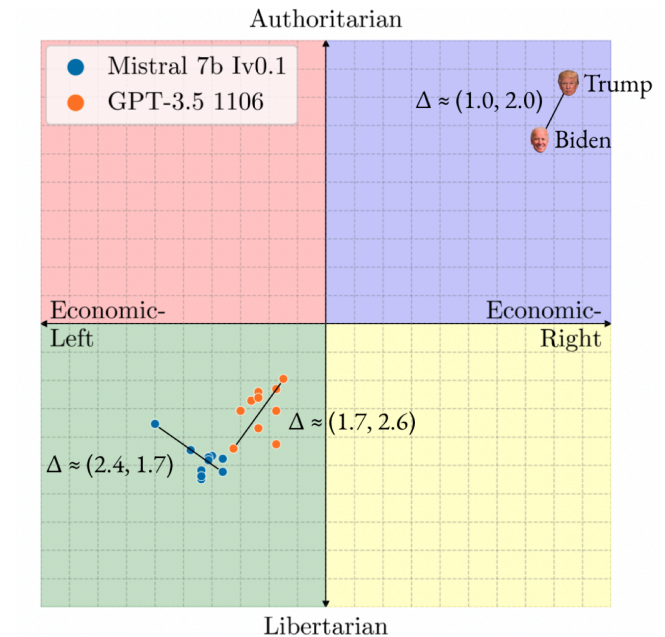
- Center-left (Ceron et al. (2024), and others)
- Social-Democrat leaning

- Cultural norms and values

- Mainly Anglo-saxon world
- W.E.I.R.D.

- Moral values and personality-like traits

- ↑ Agreeableness
- ↓ Neuroticism



Extracted from (Röttger et al. 2024).



# 5 Summary

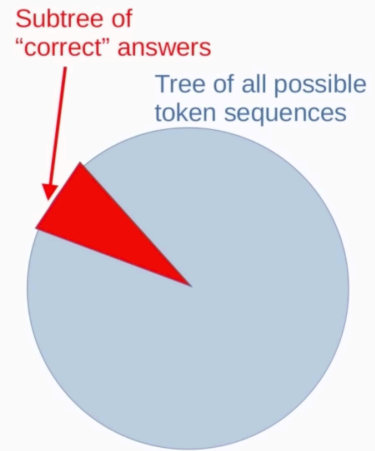
- Decoder only models trained for next-token prediction
- Pre-training data is not “partial”
- Deployed models are not only pre-trained → further aligned
- Multi-task learner and multiple use cases
- Evaluation is hard!
  - & biases run deep and have implications!

# Not the pinnacle!

## Generative Auto-Regressive Generative Models Suck!

Y. LeCun

- ▶ Auto-Regressive LLMs are **doomed**.
- ▶ They cannot be made factual, non-toxic, etc.
- ▶ They are not controllable
- ▶ Probability  $e$  that any produced token takes us outside of the set of correct answers
- ▶ Probability that answer of length  $n$  is correct (assuming independence of errors):
  - ▶  $P(\text{correct}) = (1-e)^n$
- ▶ **This diverges exponentially.**
- ▶ **It's not fixable (without a major redesign).**
- ▶ See also [Dziri...Choi, ArXiv:2305.18654]

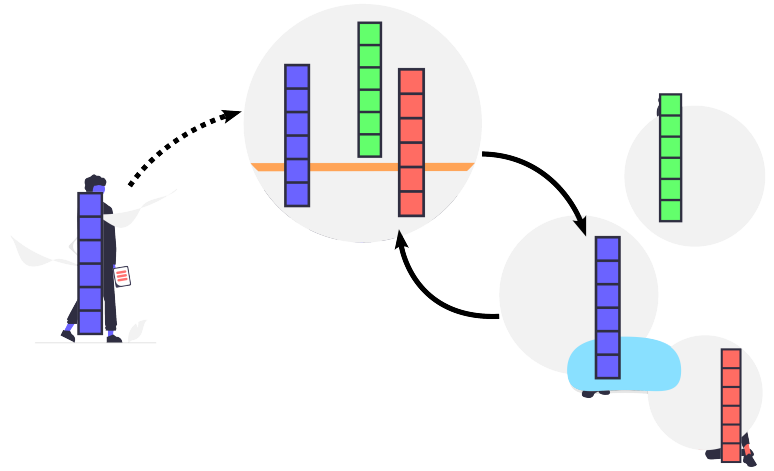
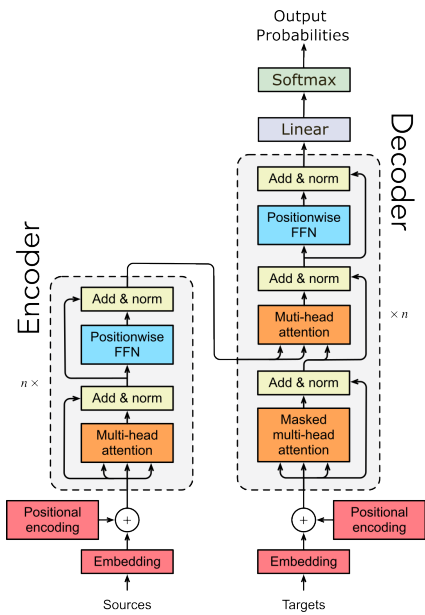


Even from a technical perspective... Slide from Yann LeCun (AMS Josiah Willard Gibbs Lecture, "Mathematical Obstacles on the Way to Human-Level AI." (2025)).  
LLM Iceberg... ©Sasha Luccioni (@SashaMTL).

**6 4 weeks / 4 slides**

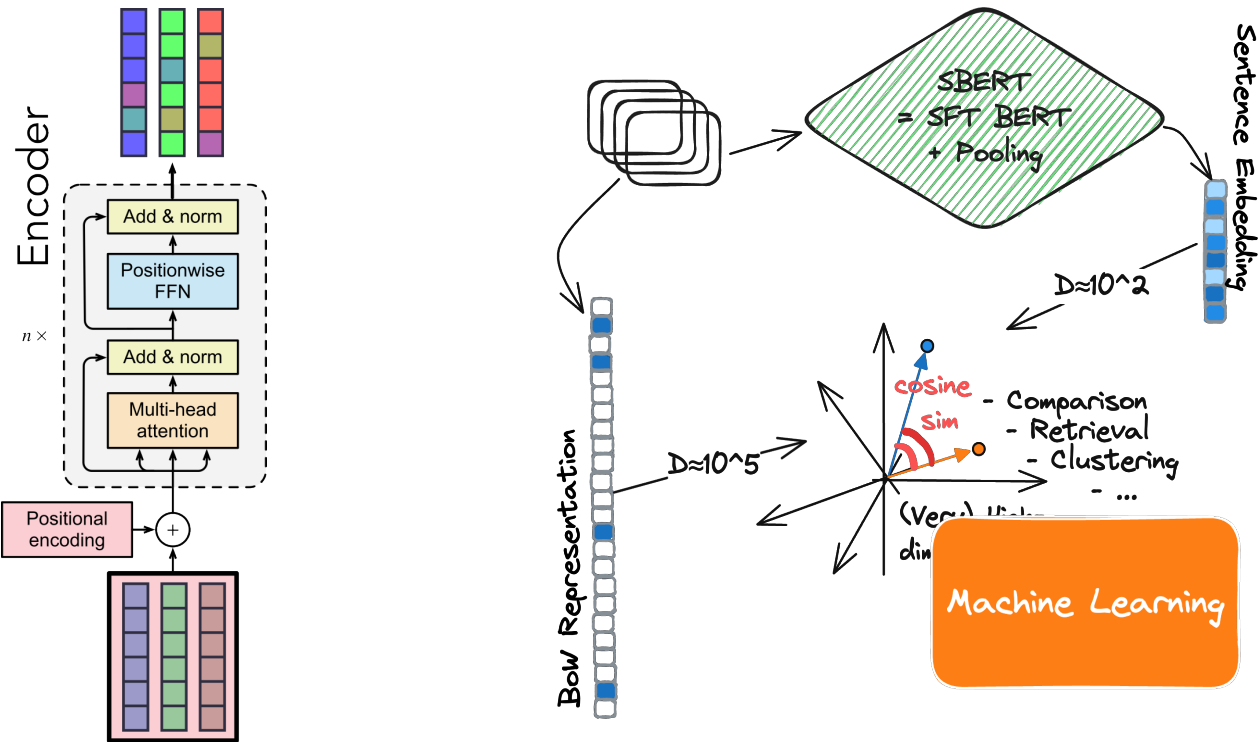
## **6.1 Language modeling and transformer-based models**

## 6.1.1 Foundations: The Transformer Architecture



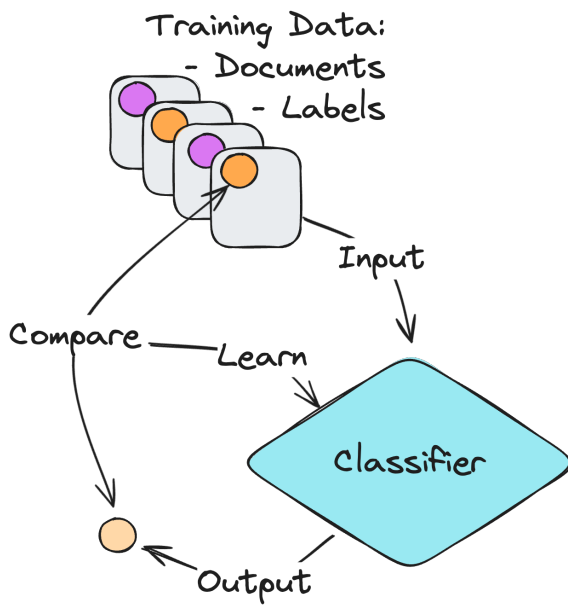
Transformer Architecture, based on (Vaswani 2017), reworked from (Zhang et al. 2023).

# 6.1.2 Feature Extraction

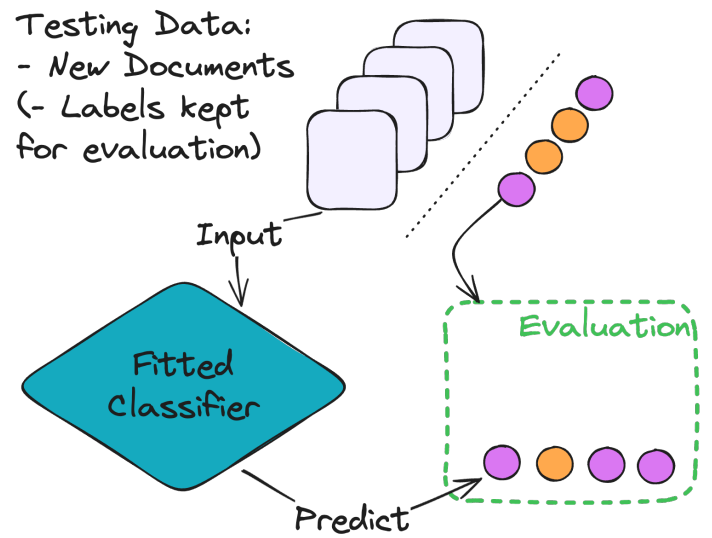


## 6.1.3 Classification and Fine-tuning

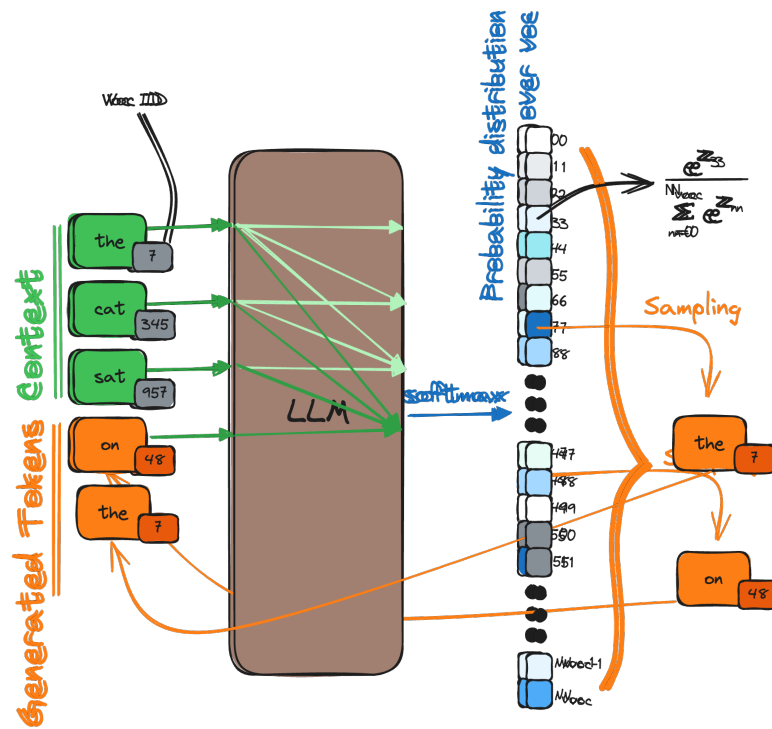
### Training Stage



### Testing Stage



### 6.1.4 Language Generation





# 6.2 In practice

## 6.2.1 Minimal (pseudo-)code for diverse tasks

### Feature Extraction

```
1 from transformers import (
2     BertTokenizer,
3     BertModel
4 )
5 docs = ["YOUR", "DOCS", "HERE"]
6 tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
7 model = BertModel.from_pretrained("bert-base-uncased")
8 tokenized_sentences = tokenizer.tokenize(docs)
9 docs,
10 truncation=True,          # Truncate inputs
11 padding=True,             # Pad inputs to max length
12 return_tensors="pt"
13 ).to(DEVICE)
14 outputs = model(**tokenized_sentences) # Run (tokenize + encode)
15 embeddings = outputs["last_hidden_state"] # Vector representations
16 # Retrieve embeddings at the position of the tokens
```

### Document Representation

- BoW / TF-IDF

```
1 from sklearn.feature_extraction.text import (
2     TfidfVectorizer,
3     CountVectorizer
4 )
5 docs = ["YOUR", "DOCS", "HERE"]
6 bow_vectorizer = CountVectorizer()
7 X_bow = bow_vectorizer.fit_transform(docs)
8 tfidf_vectorizer = TfidfVectorizer()
9 X_tfidf = vectorizer.fit_transform(docs)
```

- SentenceTransformer

```
1 from sentence_transformers import SentenceTransformer
2 docs = ["YOUR", "DOCS", "HERE"]
3 model_name = "<YOUR/MODEL_NAME>" # local or HuggingFace
4 model = SentenceTransformer(model_name)
5 embeddings = model.encode(sentences)
```

### Cosine Similarity

```
1 from sklearn.metrics.pairwise import cosine_similarity
2 docs_representations = ... # pre-computed document representations
3 pairwise_cosine_sims = cosine_similarity(docs_representations)
4 pairwise_cosine_sims[0,1] # cosine sim between docs 0 and 1
```

### Topic Modeling:

- BERTopic

```
1 from bertopic import BERTopic
2 docs = ["YOUR", "DOCS", "HERE"]
3 topic_model = BERTopic()
4 topics, probs = topic_model.fit_transform(docs)
```

- LDA

```
1 from gensim import corpora
2 from gensim.utils import simple_preprocess
3 from gensim.models import LdaModel
4 texts = ["YOUR", "DOCS", "HERE"]
5 dictionary = corpora.Dictionary(texts)
6 corpus = [dictionary.doc2bow(text) for text in texts]
7 lda_model = LdaModel(corpus=corpus, num_topics=n_topics, id2word=dictionary,
8                       corpus_preprocessed=True)
9 lda_model.fit_transform(corpus)
```

### Rep+ML (& eval)

```
1 from sklearn.metrics import classification_report
2 from sklearn.svm import SVC # or any other ML method
3 X_train, X_test = ... # pre-computed document representations
4 y_train, y_test = ... # associated labels
5 clf = SVC() # + add arguments
6 clf.fit(X_train, y_train) # train the ML method
7 y_pred = clf.predict(X_test)
8 print(classification_report(y_test, y_pred)) # or
```

### SFT (for sequence classification)

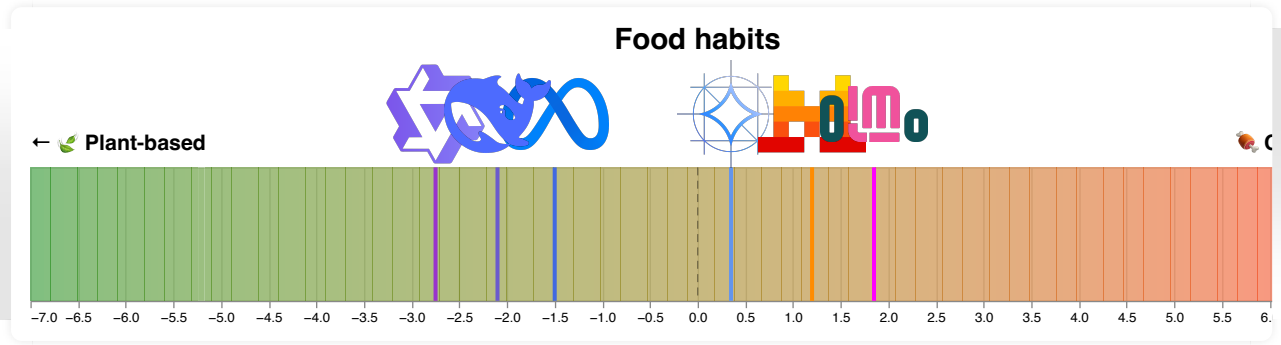
```
1 from datasets import load_dataset
2 from transformers import (
3     AutoTokenizer, AutoModelForSequenceClassification,
4     TrainingArguments, Trainer,
5 )
6
7 raw_dataset = load_dataset("<YOUR_DATASET>")
8 model_name = "YOUR_MODEL_NAME"
9 tokenizer = AutoTokenizer.from_pretrained(model_name)
10 model = AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2) # Number of labels for the classification task
11
12 def tokenize_function(example, text_column="text"):
13     return tokenizer(ex[text_column], padding="max_length", truncation=True)
14
15 training_args = TrainingArguments("test-training")
16 trainer = Trainer(
17     model=model,
18     data_loader=train_dataloader,
19     data_collator=train_data_collator,
20     tokenizer=tokenizer,
21     validation_data_loader=val_dataloader,
22     validation_data_collator=val_data_collator,
23     processing_class=tokenizer,
24 )
25 trainer.train()
```

### Generation

```
1 from transformers import pipeline
2 model_name = "<YOUR/MODEL_NAME>" # local or HuggingFace
3 prompts = ["YOUR", "PROMPTS", "HERE"] # list of prompts
4 generator = pipeline("text-generation", model=model_name, device_map="auto", max_new_tokens=max_new_tokens, batch_size=batch_size)
5
6 outputs = generator(prompts)
7
8 responses = [o[0]["generated_text"] for o in outputs]
```

# 7 Hands-on

# 7.1 Hands-On Proposition



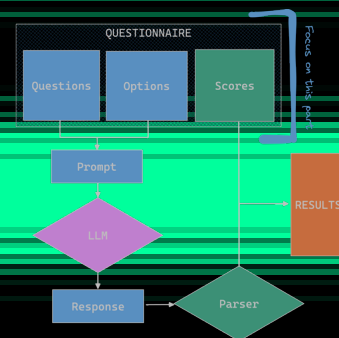
## &lt;Hands-on Session /&gt;

Machine Behavior

goal

questionnaire.json

proposed\_pipeline



reflection\_prompts

dimensions

main\_references

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